

Conjecture 00000007928 is false:
the formula $|K_2(\mathcal{O}_F)| = |\zeta_F(-1)| D_F 2^{[F:\mathbb{Q}]}$ fails for $F = \mathbb{Q}$

Jackmeson1

1 The conjecture

The English statement reads:

Definition: The spectralization of the tame kernel: a Brauer-type explicit formula for the tame kernel order $w_2(F) = |K_2(\mathcal{O}_F)|$. Conjecture: The correction to the 2-adic explicit formula $w_2(F) = \prod_p p^{\sum e(f_p)}$ is the adjustment of Dirichlet units: $w_2(F) = |\zeta_F(-1)| D_F 2^{[F:\mathbb{Q}]}$ (an integralization of a rational identity), and the p -part of the formula is given by the valuation sum of the local dilogarithm values of the tame symbol on units, with the unit contribution obeying a logarithmic law linear in the unit rank of F . (tame kernel Brauer explicit formula)

The Chinese statement has the same content. It also gives the displayed formula $w_2(F) = |K_2(\mathcal{O}_F)|$ and $w_2(F) = |\zeta_F(-1)| \cdot D_F \cdot 2^{[F:\mathbb{Q}]}$, with the same parenthetical (“integralization of a rational identity”).

2 Reading

The Definition line fixes $w_2(F) := |K_2(\mathcal{O}_F)|$, the order of the group K_2 of the ring of integers $\mathcal{O}_F$ of a number field F . The main clause asserts the equality

$$(*) \quad |K_2(\mathcal{O}_F)| = |\zeta_F(-1)| \cdot D_F \cdot 2^{[F:\mathbb{Q}]},$$

for number fields F (the quantifier is implicit; the formula is stated for a general F). The conjecture is a conjunction of $(*)$ with further clauses (the p -part given by local dilogarithm values, and a logarithmic law for the unit contribution). We refute $(*)$, which refutes the conjunction. We do not address the other clauses. Our witness is $F = \mathbb{Q}$, a totally real field whose unit rank is 0. A reading that restricts $(*)$ to fields of positive unit rank is not refuted here.

Conventions.

- ζ_F is the Dedekind zeta function. It is the Dirichlet series $\sum_{n \geq 1} a_n n^{-s}$, where a_n is the number of ideals of $\mathcal{O}_F$ of absolute norm n . This series converges for $\operatorname{Re} s > 1$. It has a meromorphic continuation to $\mathbb{C}$ whose only pole is a simple pole at $s = 1$ (Wikipedia, “Dedekind zeta function”, quoted in the attempt record). $\zeta_F(-1)$ means the value of this continuation.
- D_F is the discriminant of F , taken in absolute value $|d_F|$. For $F = \mathbb{Q}$ we have $d_{\mathbb{Q}} = 1$, so the choice of sign convention does not matter.
- $[F : \mathbb{Q}]$ is the degree $\dim_{\mathbb{Q}} F$; it equals 1 for $F = \mathbb{Q}$.
- $|G|$ is the number of elements of the group G . In Lean we use `Nat.card G`, which is 0 for infinite G . An infinite group has no finite order, so in particular it does not have order $1/6$.

The parenthetical “an integralization of a rational identity”. The main clause displays an equality between $w_2(F)$, the order of a group, and the number $|\zeta_F(-1)|D_F 2^{[F:\mathbb{Q}]}$. The parenthetical describes this equality as an integral form of a rational identity. It does not replace the displayed equality by a different formula, and it names no integralization procedure. We refute the displayed equality, which is the clause’s assertion. For $F = \mathbb{Q}$ the right-hand side is the non-integer $1/6$ (Theorem 3). So $(*)$ is not an integral identity at $F = \mathbb{Q}$, and no integral identity is obtained by asserting it there. A different reading, in which the left side equals the right side only after some unspecified operation (for example a numerator or a rounding), is a different formula. It is not refuted here.

Why K_2 itself is not built. The refutation does not use any property of $K_2(\mathbb{Z})$. The order of any group is a natural number, or else the group is infinite. The right-hand side of $(*)$ at $F = \mathbb{Q}$ is $1/6$, which is neither. The Lean theorem therefore quantifies over *every* type G and states that $\text{Nat.card } G$ is not the right-hand side. This covers $G = K_2(\mathbb{Z})$ whatever its construction. The same argument covers w_2 read in its other usual sense (a positive integer defined through roots of unity), since that is a natural number too.

Context (not used in the proof). The formula $(*)$ is not a rewriting of the Birch–Tate conjecture. Wikipedia’s Birch–Tate article, retrieved for this report, says that K_2 of a number field “is also known as the tame kernel” and that the conjecture “relates the order of this group (its number of elements) to the value of the Dedekind zeta function”. To be precise, the finite group meant here is $K_2(\mathcal{O}_F)$, the tame kernel, i.e. the kernel of the tame-symbol map from $K_2(F)$ to $\bigoplus_{\mathfrak{p}} k(\mathfrak{p})^\times$ over the finite primes; it is not $K_2(F)$ itself. The formula given there, for totally real F , is $\#K_2(\mathcal{O}_F) = |N \cdot \zeta_F(-1)|$. Here N is a positive integer defined through roots of unity. The formula $(*)$ has the factor $D_F 2^{[F:\mathbb{Q}]}$ in place of N .

3 Proof

Write $\mathcal{O}_{\mathbb{Q}}$ for the ring of integers of $\mathbb{Q}$, which is identified with $\mathbb{Z}$ by Mathlib’s `Rat.ringOfIntegersEquiv`.

Lemma 1 (`ideal_eq_span_absNorm`, `absNorm_span_nat`, `card_ideals_of_norm`). *Every ideal I of $\mathcal{O}_{\mathbb{Q}}$ equals $(N(I))$, the ideal generated by the natural number $N(I)$, its absolute norm. For each $m \in \mathbb{N}$ the ideal (m) has norm m . Hence for every n there is exactly one ideal of norm n , so $a_n = 1$.*

Proof. $\mathcal{O}_{\mathbb{Q}} \cong \mathbb{Z}$ is a principal ideal ring, so $I = (k)$ for an integer k . The norm of (k) is $|N_{\mathcal{O}_{\mathbb{Q}}/\mathbb{Z}}(k)| = |k|^{[\mathbb{Q}:\mathbb{Q}]} = |k|$, and $(k) = (|k|)$ since $k = \pm|k|$. The same norm computation gives $N((m)) = m$. Two ideals of norm n both equal (n) . $\square$

Lemma 2 (`dedekindZeta_rat`, `continuation_value`, `riemannZeta_is_continuation`). *For $\text{Re } s > 1$ the Dirichlet series of $\zeta_{\mathbb{Q}}$ equals the Riemann zeta function $\zeta(s)$. Every function Z that is holomorphic on $\mathbb{C} \setminus \{1\}$ and agrees with this series on $\text{Re } s > 1$ satisfies $Z(-1) = \zeta(-1) = -1/12$. The Riemann zeta function itself is such a Z .*

Proof. Since $a_n = 1$, the series is $\sum_{n \geq 1} n^{-s} = \zeta(s)$ (Mathlib’s `LSeries_one_eq_riemannZeta`). Both Z and ζ are analytic on the connected open set $\mathbb{C} \setminus \{1\}$, and they agree on the open half-plane $\text{Re } s > 1$. By the identity theorem they agree on $\mathbb{C} \setminus \{1\}$, which contains -1 . Finally,

Mathlib's `riemannZeta_neg_nat_eq_bernoulli` with $k = 1$ gives $\zeta(-1) = (-1)^1 B_2/2$, and $B_2 = 1/6$ (`bernoulli_two`). So $\zeta(-1) = -1/12$ (`riemannZeta_neg_one`). $\square$

Theorem 3 (`conjecture_7928_false`). *Let Z be as in the previous lemma, so $Z(-1) = \zeta_{\mathbb{Q}}(-1)$. Then*

$$|Z(-1)| \cdot |d_{\mathbb{Q}}| \cdot 2^{[\mathbb{Q}:\mathbb{Q}]} = \frac{1}{12} \cdot 1 \cdot 2 = \frac{1}{6},$$

and for every type G , $\text{Nat.card } G \neq 1/6$. In particular $()$ fails for $F = \mathbb{Q}$, whatever the group $K_2(\mathbb{Z})$ is.*

Proof. $d_{\mathbb{Q}} = 1$ (Mathlib's `Rat.numberField_discr`) and $[\mathbb{Q}:\mathbb{Q}] = 1$ (`Module.finrank_self`), so the right-hand side is $1/6$ (`rhsQ_riemannZeta`). A natural number n is either 0 or at least 1, so $n \neq 1/6$ (`nat_ne_rhsQ`). $\square$

4 Lean formalization

The package `lean/` (Lean 4, Mathlib v4.33.1) contains `Conjecture7928/Basic.lean` (141 lines, namespace `C7928`). Its only import is `import Mathlib`.

Faithfulness of $\zeta_{\mathbb{Q}}$. Mathlib's `NumberField.dedekindZeta K s` is defined as the raw L-series `LSeries (fun n => Nat.card {I : Ideal (O K) // absNorm I = n}) s`. At $s = -1$ this series diverges, so its Lean value there is a junk value and not $\zeta_{\mathbb{Q}}(-1)$. We therefore do not evaluate it at -1 . We prove that it equals `riemannZeta` on $\text{Re } s > 1$. The main theorem then takes *any* function Z holomorphic on $\mathbb{C} \setminus \{1\}$ that agrees with `dedekindZeta  $\mathbb{Q}$  s` on $\text{Re } s > 1$, that is, the continuation, and evaluates $Z(-1)$. The hypotheses are not vacuous: `riemannZeta` satisfies them (`riemannZeta_is_continuation`).

Main statements.

```
theorem conjecture_7928_false (Z : C -> C) (hZ : DifferentiableOn C Z {1}^c)
  (hZeq : forall s : C, 1 < s.re -> Z s = dedekindZeta Q s) (G : Type*) :
  (Nat.card G : R) !=
    ||Z (-1)|| * |((NumberField.discr Q : Z) : R)| * 2 ^ Module.finrank Q Q
```

```
theorem conjecture_7928_false_riemannZeta (G : Type*) :
  (Nat.card G : R) !=
    ||riemannZeta (-1)|| * |((NumberField.discr Q : Z) : R)|
    * 2 ^ Module.finrank Q Q
```

```
theorem conjecture_7928_false_finite_group (G : Type*) [Group G] [Fintype G] :
  (Fintype.card G : R) != (same right-hand side)
```

(ASCII rendering: $\mathbb{C}, \mathbb{R}, \mathbb{Q}, \mathbb{Z}$ stand for $\mathbb{C}, \mathbb{R}, \mathbb{Q}, \mathbb{Z}$; $!=$ is $\neq$; $||\cdot||$ is the complex norm.) Supporting lemmas: `riemannZeta_neg_one`, `rhsQ_riemannZeta`, `ideal_eq_span_absNorm`, `absNorm_span_nat`, `card_ideals_of_norm`, `dedekindZeta_rat`, `continuation_value`, `riemannZeta_is_continuation`, `nat_ne_rhsQ`. The identity-theorem step follows the pattern of Mathlib's own proof in `Mathlib/NumberTheory/Harmonic/ZetaAsymp.lean`.

Axiom audit. `lake build` succeeds. `#print` axioms for each of `conjecture_7928_false`, `conjecture_7928_false_riemannZeta`, `conjecture_7928_false_finite_group` and `riemannZeta_is_continuation` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry`, no `native_decide` and no added axiom.

Sources

- Wikipedia, “Dedekind zeta function”, https://en.wikipedia.org/wiki/Dedekind_zeta_function (Dirichlet series and meromorphic continuation with only a simple pole at $s = 1$).
- Wikipedia, “Birch–Tate conjecture”, https://en.wikipedia.org/wiki/Birch%E2%80%93Tate_conjecture (context only).